

DATA ANALYTICS LEVEL III

Day 5: SQL for Data Preparation — Import & Filter Data

Student Handout

Learning Objectives

By the end of Day 5, you should be able to:

- Import external data (CSV) into a SQLite database
- Use WHERE with multiple conditions (AND, OR, NOT)
- Filter data using BETWEEN, IN, and LIKE operators
- Sort query results with ORDER BY (ASC and DESC)
- Limit output with LIMIT
- Apply SQL filtering to answer real-world business questions

Core Competency: Core 2 — Prepare Data Sets

21st Century Skills: Critical thinking, Collaboration, Digital literacy

Recap: What We Learned Yesterday

- A database is an organized, structured collection of data
- SQL (Structured Query Language) is the standard language for databases
- SELECT chooses columns, FROM specifies the table, WHERE filters rows
- Text values need single quotes; numbers do not

Today we go much deeper into filtering — the most important skill for data preparation.

Importing Data into SQL

Why Import?

In real work, data often starts as CSV files, Excel exports, or other flat files. You need to bring that data INTO the database before you can query it.

Method 1: CREATE TABLE + INSERT (Manual)

You can create a table and insert data row by row:

```
CREATE TABLE new_customers (
  customer_id INTEGER PRIMARY KEY,
  first_name TEXT NOT NULL,
  last_name TEXT NOT NULL,
  email TEXT NOT NULL,
  city TEXT NOT NULL
);
```

```
INSERT INTO new_customers VALUES (1, 'Lily', 'Chua', 'lily@email.com', 'Manila');
INSERT INTO new_customers VALUES (2, 'Mark', 'Reyes', 'mark@email.com', 'Cebu City');
```

This works for small amounts of data but is tedious for large files.

Method 2: Import CSV in VS Code / SQLite Tools

For larger files, you can use the SQLite command-line or tools to import CSVs:

```
-- In SQLite command-line mode:
.mode csv
.import Day05_new_customers.csv customers
```

In VS Code with SQLTools, you may need to use INSERT statements or a script. Your instructor will demonstrate the process.

Key Points About Importing

- Column names in the CSV must match the table's column names
- Data types must be compatible (text in text columns, numbers in number columns)
- Watch for extra spaces, special characters, and encoding issues
- **Always verify your import:** `SELECT COUNT(*) FROM table_name;`

Advanced WHERE: Multiple Conditions

AND — Both Conditions Must Be True

Use AND when you need rows that satisfy ALL conditions:

```
SELECT * FROM sales
WHERE region = 'NCR' AND total_amount > 10000;
```

```
-- NCR sales above ₱10,000
```

OR — At Least One Condition Must Be True

Use OR when you need rows that satisfy ANY condition:

```
SELECT * FROM sales
```

```
WHERE city = 'Manila' OR city = 'Makati';
```

```
-- Sales from Manila OR Makati
```

NOT — Reverses a Condition

Use NOT to exclude matching rows:

```
SELECT * FROM sales
WHERE NOT category = 'Accessories';
```

```
-- Everything EXCEPT Accessories
```

Combining AND, OR, NOT

You can combine these, but use parentheses to control the order:

```
SELECT * FROM sales
WHERE (city = 'Manila' OR city = 'Makati')
    AND total_amount > 5000;
```

```
-- Manila or Makati sales above ₱5,000
```

Important: Without parentheses, AND is evaluated before OR, which can give unexpected results. Always use parentheses when mixing AND and OR.

BETWEEN, IN, and LIKE

BETWEEN — Range Filter

Filters values within a range (inclusive on both ends):

```
SELECT * FROM sales
WHERE total_amount BETWEEN 1000 AND 5000;
```

```
-- Sales from ₱1,000 to ₱5,000 (inclusive)
```

Also works with dates:

```
SELECT * FROM sales
WHERE sale_date BETWEEN '2025-01-01' AND '2025-03-31';
```

```
-- All Q1 2025 sales
```

IN — Match Any Value in a List

A cleaner way to check multiple values (instead of many ORs):

```
-- Instead of this:
SELECT * FROM sales
WHERE city = 'Manila' OR city = 'Makati' OR city = 'Pasig';
```

```
-- Use this:
SELECT * FROM sales
WHERE city IN ('Manila', 'Makati', 'Pasig');
```

LIKE — Pattern Matching

Searches for patterns in text columns using wildcards:

Wildcard	Meaning	Example	Matches
%	Any sequence of characters (0 or more)	'M%'	Manila, Makati, Miguel
_	Exactly one character	'_a%'	Dan, Patricia (2nd letter is 'a')

```
-- Products that start with 'Laptop'
SELECT * FROM sales
WHERE product_name LIKE 'Laptop%';
```

```
-- Products that contain 'USB'
SELECT * FROM sales
WHERE product_name LIKE '%USB%';
```

```
-- Cities that end with 'City'
SELECT * FROM sales
WHERE city LIKE '%City';
```

Note: In SQLite, LIKE is case-insensitive by default for ASCII characters.

ORDER BY — Sorting Results

ORDER BY sorts your results. Think of it as the ‘Sort’ function in Excel.

```
-- Sort by total_amount, lowest to highest (ascending)
SELECT * FROM sales
ORDER BY total_amount ASC;
```

```
-- Sort by total_amount, highest to lowest (descending)
SELECT * FROM sales
ORDER BY total_amount DESC;
```

```
-- Sort by multiple columns
SELECT * FROM sales
ORDER BY category ASC, total_amount DESC;
```

ASC = ascending (A→Z, 1→100) — this is the default if you don’t specify.
DESC = descending (Z→A, 100→1)

LIMIT — Controlling How Many Rows You See

LIMIT restricts the number of rows returned:

```
-- Show only the top 5 highest sales
SELECT * FROM sales
ORDER BY total_amount DESC
LIMIT 5;
```

```
-- Show the 10 most recent sales
SELECT * FROM sales
ORDER BY sale_date DESC
LIMIT 10;
```

LIMIT is especially useful for large tables when you want a quick preview: `SELECT * FROM sales LIMIT 10;`

Putting It All Together: SQL Query Order

SQL clauses must appear in this specific order:

```
SELECT columns
FROM table
WHERE conditions
ORDER BY column ASC/DESC
LIMIT number;
```

Think of it as: “What do I want? From where? With what filter? In what order? How many?”

Example — Top 5 highest NCR sales of Electronics:

```
SELECT customer_name, product_name, total_amount
FROM sales
WHERE region = 'NCR' AND category = 'Electronics'
ORDER BY total_amount DESC
LIMIT 5;
```

Practice Database: What’s Inside?

Your Day05_practice.db contains three tables:

sales table (50 rows)

Column	Type	Description
sale_id	Integer	Unique ID
sale_date	Text	Date of sale (YYYY-MM-DD)
customer_name	Text	Buyer’s name

city	Text	Buyer's city
region	Text	NCR, Visayas, or Mindanao
product_name	Text	Item purchased
category	Text	Electronics, Accessories, Storage
quantity	Integer	Units bought
unit_price	Real	Price per unit (₱)
total_amount	Real	Total sale (₱)
payment_method	Text	GCash, Credit Card, Cash

inventory table (25 rows)

Column	Type	Description
item_id	Integer	Unique ID
item_name	Text	Product name
category	Text	Electronics, Accessories, Storage
warehouse	Text	Manila/Cebu/Davao Warehouse
quantity_on_hand	Integer	Current stock
reorder_level	Integer	Minimum before reorder
unit_cost	Real	Cost per unit (₱)
supplier	Text	Supplier name
last_restock_date	Text	Last restock date

customers table (20 rows)

Column	Type	Description
customer_id	Integer	Unique ID
first_name	Text	First name
last_name	Text	Last name
email	Text	Email address
city	Text	City
region	Text	NCR, Visayas, Mindanao
signup_date	Text	Registration date
total_orders	Integer	Lifetime order count

total_spent	Real	Lifetime spending (₱)
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SQL Quick Reference

Concept	Syntax	Example
Multiple conditions (all)	WHERE x AND y	WHERE region='NCR' AND amount>5000
Multiple conditions (any)	WHERE x OR y	WHERE city='Manila' OR city='Cebu City'
Exclude	WHERE NOT x	WHERE NOT category='Accessories'
Range	WHERE col BETWEEN a AND b	WHERE amount BETWEEN 1000 AND 5000
List match	WHERE col IN (list)	WHERE city IN ('Manila', 'Makati')
Pattern	WHERE col LIKE pattern	WHERE name LIKE 'L%'
Sort ascending	ORDER BY col ASC	ORDER BY total_amount ASC
Sort descending	ORDER BY col DESC	ORDER BY total_amount DESC
Limit rows	LIMIT n	LIMIT 10

Study Activities

Your Day 5 activities are in a separate document: Day05_Student_Activities.docx (download from LMS).

Today's asynchronous tasks (3.5 hours total):

- **Reading:** Review this Student Handout — especially BETWEEN, IN, LIKE sections (45 minutes)
- **SQL Practice:** Filter & Sort Exercises — 15 progressive queries (1.5 hours)
- **Activity:** Real-World Query Scenarios — 5 business questions (45 minutes)
- Learning Journal Entry #5 (30 minutes)

All submissions are due before the next session via the LMS.

Key Takeaways

- AND requires ALL conditions to be true; OR requires at least one
- Always use parentheses when combining AND and OR

- BETWEEN filters ranges (inclusive); IN matches a list of values
 - LIKE with % and _ wildcards enables pattern matching
 - ORDER BY sorts results; LIMIT controls how many rows you see
 - **SQL query order:** SELECT → FROM → WHERE → ORDER BY → LIMIT
 - Always verify imported data with SELECT COUNT(*)
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Notes

Use this space for your own notes:
